

Applied A.I. Solutions

Agile Project Management Methodologies

Quotations, Reference and Recommended Bibliography:

- 1 - "The Epic Guide to Agile – More Business Value On A Predictable Schedule With Scrum" – Author: Dave Todaro - Copyright © 2019 David M. Todaro – Published by R9 Publishing LLC
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Agile Project Management Methodologies

Ritual + Rhythm + Communication + Value +
Commitment + Teamwork + Discipline + Transparency
+ Continuous Improvement + Rapid Development

Week 1 / Day 1

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Introduction

- Nearly 20 years ago, 17 software developers came together in Snowbird, Utah to propose a new way of developing software “by doing it and helping others do it.”
- The signers of the Manifesto understood how much of an impact these principles would help them in the field of software development—but they had no idea how quickly their ideas would spread beyond their industry.
- 4 Values:

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AGILE MANIFESTO

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

Kent Beck
Mike Beedle
Arie van Bennekum
Alistair Cockburn
Ward Cunningham
Martin Fowler

James Grenning
Jim Highsmith
Andrew Hunt
Ron Jeffries
Jon Kern
Brian Marick

Robert C. Martin
Steve Mellor
Ken Schwaber
Jeff Sutherland
Dave Thomas

<http://agilemanifesto.org/>

<https://www.scrumalliance.org/resources/agile-manifesto>

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Key principles for incremental development

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

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Key principles for incremental development – cont'd

4. Business-people and developers must **work together daily** throughout the project.
5. Build projects around **motivated individuals**. Give them the environment and support they need and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is **face-to-face conversation**.

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Key principles for incremental development – cont'd

7. Working software is the primary measure of progress.
What does that mean? what does 90% complete mean?
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.

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Key principles for incremental development – cont'd

10. **Simplicity**--the art of maximizing the amount of work not done--is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

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What is Agile methodology?

- Agile methodology is a team-based approach that emphasizes rapid deployment of a functional application with a focus on customer satisfaction.
- It defines a time-boxed phase called a sprint with a defined duration of two weeks.
- At the start of each sprint, a list of deliverables are prioritized based on customer input.

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What is Agile methodology? – cont'd

- At the end of the sprint, the team and the customer review and evaluate the work with notes for future sprints.
- As a methodology based on general principles, more specific methods based on processes such as **Scrum and Kanban** are called types of agile methodology.

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What are the benefits of Agile?

- Faster software development life cycle
- Predictable schedule in sprints
- Customer-focused, resulting in increased customer satisfaction
- Flexible in accepting changes
- Empowers teams to manage projects
- Promotes efficient communications
- Ideal for projects with non-fixed funding

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What are the disadvantages of Agile?

- Agile requires a high degree of customer **involvement**, which not all customers are comfortable with or prefer to give.
- Agile assumes every team member is completely **dedicated** to the project, without which weakens the principle of self-management.
- A time-boxed iteration may not be enough to accommodate all deliverables, which will require changes in **priority** and additional sprints that can bring up cost.
- Agile recommends **co-location** for efficient communication, which is not always possible.

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What is Waterfall methodology?

- Waterfall methodology is a sequential approach that divides the SDLC to distinct phases such as requirements gathering, analysis and design, coding and unit testing, system and user acceptance testing, and deployment.

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What is Waterfall methodology? – cont'd

- All phases are passed through and completed only once, so all requirements are gathered as much as possible at the start to provide the information in creating the plans, schedules, budget, and resources.
- It is plan-driven, so any changes after the project has started would offset the original plan and require a restart.

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What are the benefits of Waterfall?

- Straightforward planning and designing due to the agreement on deliverables at the start of the project.
- Better design with whole-system approach
- Defined scope of work
- Easier costing
- Clear measurements of progress
- Defined team roles, dedicated resources can work in parallel for their specific tasks

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What are the disadvantages of Waterfall?

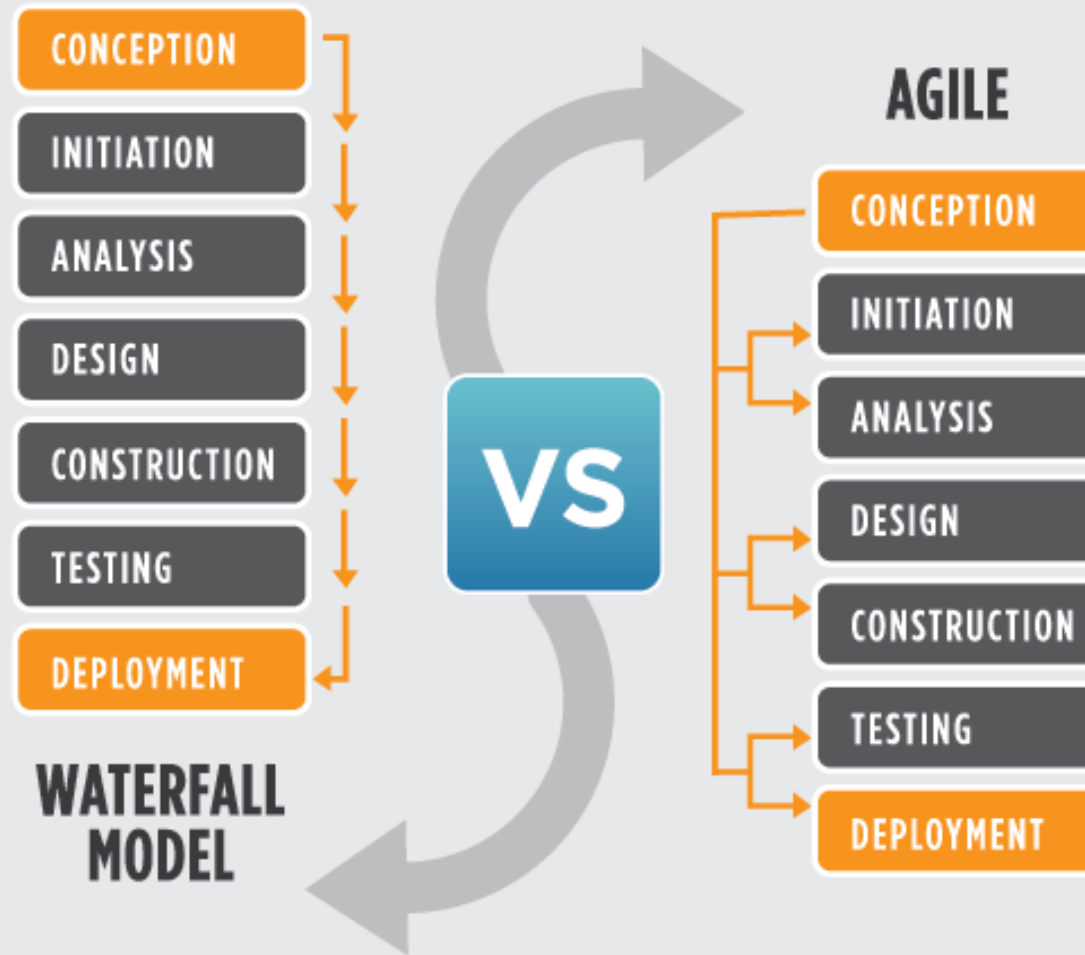
- Rigid structure to allow necessary changes
- No allowance for uncertainty
- Limited customer engagement, resulting in poor satisfaction
- Sequential approach is not ideal for a large-sized project where the end result is too far in the future
- Testing is done only at the latter phases of the project.

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ILLUSTRATED BY SEGUE TECHNOLOGIES

Waterfall Project Management	Agile Scrum Project Management
project mindset with a focus on successful project delivery	product mindset with a focus on customer satisfaction
linear and sequential approach	incremental and iterative approach
divides a project into phases	separates a project into sprints
helps complete one single project	helps complete many small projects
requirements are prepared once at the start	requirements are prepared everyday
avoids scope changes once the project starts	allows requirement changes at any time
testing phase comes only after the build phase	testing is performed concurrently with development
test teams do not get involved in requirements change	Test teams can take part in requirements change
requires a project manager who plays an essential role in every phase	enables the entire team to manage the project without a dedicated project manager

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High-performance Development Team

- Delivers the highest value to the business
- Gets an accurate schedule forecast to rely on
- Take ownership and accountability
- Get a high-quality product that users love

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How to achieve a high-performance team?

- Help the team to understand business goals, effectively communicate with each other, coordinate their work on a way that drives the best result.
- Help the team to predict how long it will take to complete their work and ensure that they consistently produce the same high-quality results.
- Help the team to improve over time so they can keep getting better at what they do.

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Scrum

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What is Scrum?

- It is a flexible and adaptable tool that allows a group of individuals become a high-performance team.
- Scrum provides a way for business leaders to drive priorities and have a development team produce results consistently, reliably, and predictably.
- It works because it is **social engineering framework**. By imposing a small set of rules, it allows a group of normal individuals to do the right things, at the right time, to produce extraordinary results.

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What is a sprint?

Work is completed in **fixed-duration production** cycles called “**sprints**” or iterations.

This time period – most commonly 2 weeks – (but ranging between 1 week to a calendar month) – turns the traditional long-term deadline on its head and **forces the Scum team to produce results NOW!**

Thus, Scrum provides the framework to develop a plan and gives the team an **imminent deadline** to produce results at the end of each sprint.

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Scrum

- Forces a team to make a plan
- Doesn't provide a team quite enough time
- Forces a team to communicate
- Forces a team to be held accountable

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- Scrum was created in the early '90s by Jeff Sutherland and Ken Schwaber.

“It is a framework within which people can address complex adaptive problems, while productively and creatively delivering products of the highest possible value.”

- Scrum is especially helpful in software development projects, and/or for managing projects that have the following characteristics:
 - Emergent requirements
 - Capabilities don't necessarily exist
 - Unforeseeable uncertainty

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- **Scrum IS NOT for every project.** For example, if you are building a new office building, using traditional construction techniques, you probably don't need Scrum.
- However, if you have requirements that are likely to **evolve** over the course of the project, and at least you have some technical **unknowns** about how exactly the work will be completed , Scrum will likely may your life easier.
- Scrum is for **any** complicated or complex project.
- Scrum substantially reduces timeframes by forcing all work to be done during each fixed-production cycle or sprint.

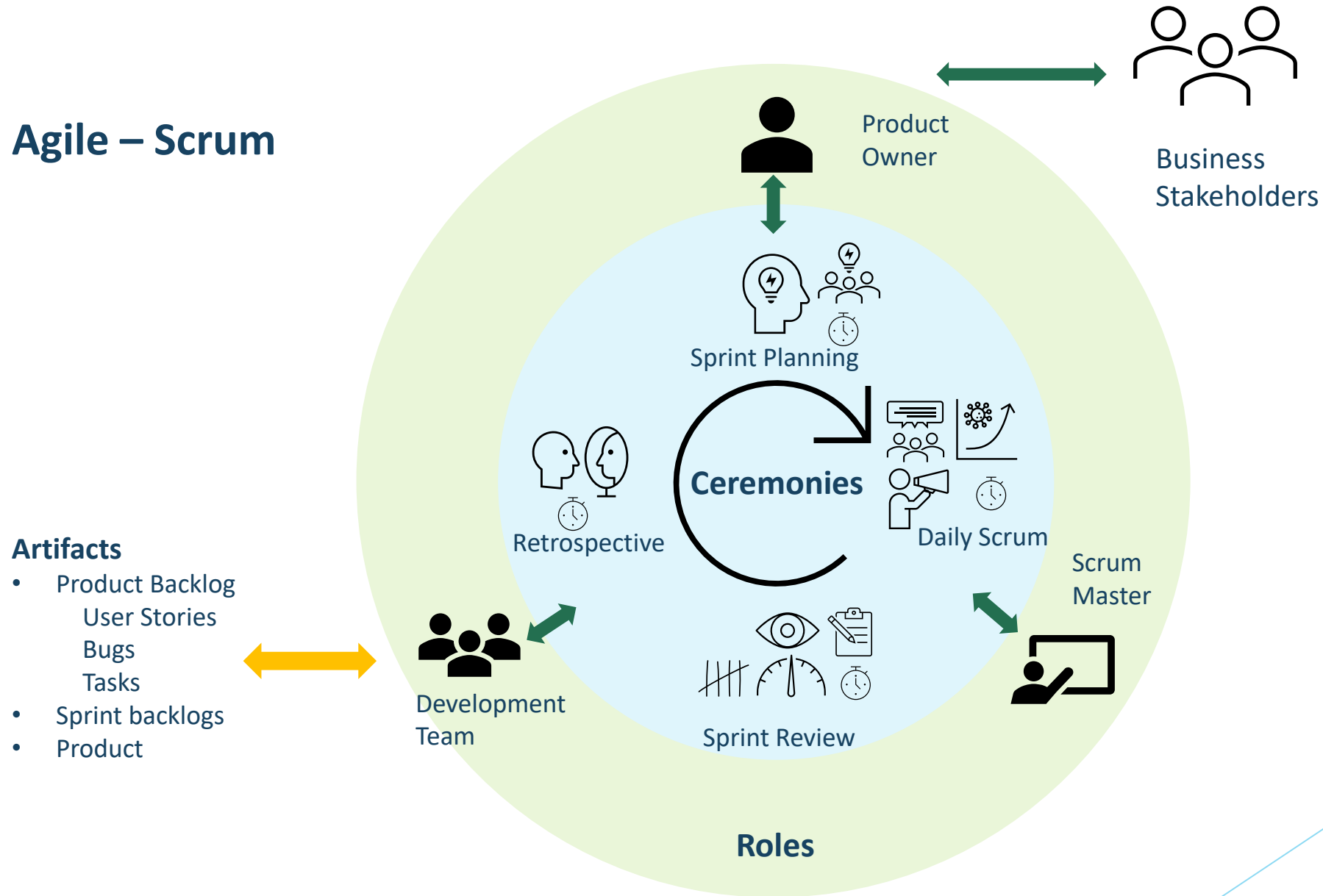
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Agile – Scrum



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Scrum: Three Roles

1. Product Owner

- Responsible for the overall vision of the application being built (product)
- Responsible for what the development team will work on, and in what order.
- Represent the business within the team, as they are responsible for distilling the views of all the various business stakeholders and carrying that views into the Scrum team.

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Scrum: Three Roles

1. **Product Owner** responsibilities include:

- Create and maintain **product backlog items**.
- **Prioritize** the product backlog to maximize business value
- Ensure product backlog items are clear, and answer questions as necessary to ensure everyone on the Scrum team understands them.
- Work closely with development team to adjust product backlog items as necessary to maximize value as the product is built. For example: adjusting acceptance criteria to make a user story easier and faster to build.
- **Review and accept or reject the work of the development team.**

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Scrum: Three Roles – cont'd

2. Scrum Master

- Responsible for the process.
- Represents the Scrum Coach on the team.
- Helps everyone understand and adhere to the fundamental Scrum concepts, and Agile values and principles.
- Protects the team from outside influence to avoid distractions.
- Helps the team resolve impediments to their progress and reflect on how well the process is running and helps facilitate improvements.

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Scrum: Three Roles – cont'd

2. Scrum Master responsibilities in service to the Product Owner

- Ensures the vision of the Product Owner, including goals, scope, and context of the product, are understood by everyone on the Scrum team.
- Help the Product Owner with the Product Backlog, experimenting with new ideas for managing effectively.
- Educate the Scrum team about why it is important that the product backlog is clear and concise.
- Help the Product Owner and Development Team understand the estimating and release planning process utilizing user stories, story points estimates, and velocity.

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Scrum: Three Roles – cont'd

2. Scrum Master responsibilities in service to the Product Owner – cont'd

- Ensures the Product Owner knows how to prioritize the product backlog to maximize value.
- Assist the Product Owner and development team with understanding how user stories, bugs, and tasks are utilized on the product backlog.
- Facilitate Scrum ceremonies.

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Scrum: Three Roles – cont'd

2. Scrum Master responsibilities **in service to the Development Team**

- **Help** the development team understand the Scrum framework and ensure they are working within the guideposts
- **Coach the development team on how to become self-organized and cross-functional**
- **Ensure effective communication** between the product Owner and the development team.
- **Remove impediments** to the development team's progress.

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Scrum: Three Roles – cont'd

2. Scrum Master responsibilities in service to the Development Team – cont'd

- **Facilitate Scrum ceremonies** and ensure each development team member is engaged and participating during each ceremony.
- **Protect the development** team from outside distractions and an overly aggressive Product Owner.
- **Ensure** the development team continually innovates and improves the process

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Scrum: Three Roles – cont'd

2. Scrum Master responsibilities in service to the Organization

- Help the organization in its Scrum adoption.
- Support Scrum implementations within the organization.
- Facilitates change that increase productivity of the Scrum team
- Work with other Scrum masters to increase effectiveness
- Help those not on the Scrum team understand which interactions with the team are helpful and those that are getting in the way of progress.

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Scrum: Three Roles – cont'd

3. Development Team

- Group of individuals who collectively possess all the required skills to complete the work.
- This group –typically comprised of seven plus or minus two individuals– does all the work on the product.
- The Product Owner and the Scrum Master are not included in this number unless they are performing work to get the product done, in which case they are also part of the development team.

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Scrum: Three Roles – cont'd

3. Development Team characteristics – cont'd

- The team is self-organizing.
- The team is cross-functional, collectively possessing all the necessary skills to create the product.
- No titles are used for members of the development team, regardless of the skills possessed by each member.

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Four Ceremonies (or events)

Ceremonies are time-boxed: they must be completed on or before a defined time to increase productivity

1. **Sprint Planning**

- They happen at the beginning of each sprint.
- The Development Team, helped by the Product Owner and Scrum Master, starts with the most important item and creates a plan to get the work done by breaking down it into subtasks, each with a time estimate in hours.
- This ceremony is time-boxed to no more than four hours for teams with two-week sprints, and eight hours for teams with a month-long sprints.

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Four Ceremonies (or events) – cont'd

2. Daily Scrum (or daily standup)

- It ensures that all members of the Development Team are communicating on a regular basis. Product Owner and Scrum Master participate if they are part of the Development Team.
- Daily Scrums are timeboxed to 15 minutes each, and occur at the same time and place, each workday during the sprint.
- Each member of the development team answers the following: what did you get done since the last daily scrum? what do you commit to getting done before the next daily scrum? what impediments are standing in our way of completing the work in the sprint?

Quotations, Reference and Recommended Bibliography:

1 - "The Epic Guide to Agile – More Business Value On A Predictable Schedule With Scrum" – Author: Dave Todaro - Copyright © 2019 David M. Todaro – Published by R9 Publishing LLC

2 - "Agile Project Management with Scrum" – Author: Ken Schwaber – Copyright © 2004 by Ken Schwaber – Published by Microsoft Press

3 - "Agile Project Management with Kanban" – Author: Eric Brechner – Copyright © 2015 by Eric Brechner – Published by Microsoft Press

Four Ceremonies (or events) – cont'd

3. Sprint Review

- At the end of each sprint to provide insight, transparency to stakeholders.
- The scrum team discusses the work of the sprint and their results.
- It includes a demonstration of the product, including any new functionality completed during the sprint and driven to a shippable level of quality.
- The scrum team discusses what is coming up next on the product backlog.

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Four Ceremonies (or events) – cont'd

3. Sprint Review – cont'd

- This ceremony is design to be a two-way interactive conversation between the scrum team and the business stakeholders.
- The ceremony is timeboxed to two hours for a two-week sprint and four hours for a month-long sprint.

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Four Ceremonies (or events) – cont'd

4. Retrospective

- It is a way for the scrum team to discuss how things went during the sprint and determine changes to improve their process.
- It is critical that each team adjusts what is inside each guidepost to what work better for them.
- This ceremony follows the sprint review and marks the end of the sprint.

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Four Ceremonies (or events) – cont'd

4. Retrospective – cont'd

- The team discuss the following questions: What went well? What could be improved? What experiments should we try next sprint to improve?
- This ceremony is time boxed to a maximum of an hour and a half for two-week sprints and three hours for month-long sprints.

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2 - "Agile Project Management with Scrum" – Author: Ken Schwaber – Copyright © 2004 by Ken Schwaber – Published by Microsoft Press

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End of Week 1 / Day 1

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